



Mitesh Kumar

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SUMMARY

Master's student in Data Science at FAU with expertise in machine learning, optimization, and computer vision. Proven track record of developing high-performance ML pipelines (90retrieval) and mathematical optimization models for complex network systems. Strong foundation in Python, PyTorch, and optimization frameworks (Pyomo, Gurobi). Seeking data science or machine learning engineering opportunities to apply quantitative methods to real-world business challenges.

WORK EXPERIENCE

UNIVERSITY OF REGENSBURG

Student Research Assistant — Machine Learning for Side-Channel Analysis

Regensburg, Germany

12/2025 – Present

- Reviewing and synthesizing literature on deep learning-based side-channel attacks, including profiling and single-trace attacks.
- Analyzing ML-assisted key-recovery attacks on post-quantum signature schemes (e.g., multivariate cryptography).
- Investigating robustness of neural-network-based attacks under realistic measurement conditions (noise, misalignment).
- Supporting the design of experimental pipelines for power-trace acquisition and ML-based leakage analysis.

FRIEDRICH-ALEXANDER-UNIVERSITÄT ERLANGEN-NÜRNBERG

Student Research Assistant — Transregio 154: Mathematical modeling, simulation and optimization using the example of gas networks.

Erlangen, Germany

05/2025 – Present

- Engineered a bicriteria optimization model in Python (Pyomo) for gas networks, successfully balancing social welfare maximization with a data-driven explainability objective by implementing dynamic, pressure-based temporal features and dynamic weighting (λ, β) .
- Scaled numerical experiments across an HPC cluster using Slurm job arrays and developed robust Min-Max normalization algorithms to accurately aggregate and visualize 50+ multi-objective Pareto fronts, successfully resolving extreme mathematical scaling distortions.
- Developed and implemented a non-convex nominal model using Pyomo and Gurobi to maximize the social welfare of a gas network subject to physical transport constraints.
- Engineered a Python workflow to generate a large-scale time series dataset by simulating and solving thousands of optimal network states.

SKILLS

Technical Skills

- Programming: Python, SQL, C++, LaTeX
- ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras
- Optimization: Pyomo, Gurobi
- Data & Visualization: Power BI, Pandas, NumPy, Matplotlib
- Tools: Git, Microsoft Office

Domain Expertise

- Machine Learning & Deep Learning
- Optimization & Mathematical Modeling
- Computer Vision

- Cryptographic Analysis

Language Expertise

- English (IELTS C1)
- Hindi (Native)
- German (Elementary)

PROJECTS

Hierarchical Object Detection via Selective Search & R-CNN Framework - Engineered object detection pipeline integrating multi-scale Selective Search, ResNet18 CNN for feature extraction, and Linear SVM for classification. - Achieved MABO of 0.74 through Non-Maximum Suppression (NMS) and geometric data augmentation, demonstrating effective region proposal quality. - Developed region proposal algorithm using color, texture, size, and fill similarities on Felzenszwalb segmentation. - Trained RBF-Kernel SVM with balanced class weighting and automated dataset balancing.	01/2025 – 02/2026
Open-Set Face Recognition & Robust Tracking System - Achieved 90% validation accuracy in open-set identification by comparing Single and Multi Pseudo Label (SPL/MPL) strategies to distinguish known users from strangers. - Optimized efficiency via prediction pipeline using dynamic K-Means clustering for unknown identities and vectorized Nearest Neighbor searches. - Built face recognition pipeline using FaceNet (ResNet50) with dual-stream color/grayscale fusion for 128-dimensional embeddings. - Integrated MTCNN for detection with Template Matching fallback tracker for identity continuity during occlusions.	12/2025 – 01/2026
Large-Scale Writer Retrieval via Multi-VLAD & Exemplar-SVM - Achieved mAP of 0.78 on ICDAR17 dataset using Upright RootSIFT descriptors and Exemplar SVMs (One-vs-Rest classification). - Boosted baseline performance by 24% via Generalized Max Pooling (GMP) and PCA-whitened Multi-VLAD ensemble for discriminative representations. - Compressed local features into global representations using VLAD with power normalization and L2 scaling. - Integrated Generalized Max Pooling with balancing parameter (λ) to weight frequent versus rare visual words.	11/2025 – 12/2025
Computational Imaging Pipeline: Demosaicing & HDR Reconstruction - Synthesized HDR images from JPG sequences by reverse-engineering camera response function through mathematical optimization, recovering linear irradiance data. - Implemented custom pipeline from scratch including Bayer demosaicing and logarithmic tone-mapping to compress dynamic ranges without clipping artifacts. - Developed HDR reconstruction merging multiple exposures into radiance maps, preserving shadow and highlight detail. - Applied tone mapping via iCAM06 model and local contrast enhancement for HDR display.	10/2025 – 11/2025
3D Scene Perception & Robust Box Dimension Estimation - Extracted precise 3D dimensions from noisy ToF sensor data using MLESAC plane-fitting and morphological filters to eliminate background noise. - Optimized for real-time constraints via custom Preemptive RANSAC strategy, reducing computational overhead while maintaining geometric accuracy. - Implemented RANSAC variants (MLESAC, Preemptive RANSAC) for plane segmentation with sensor noise handling. - Automated object isolation via Connected Component Analysis, mapping 2D coordinates to 3D Euclidean space.	09/2025 – 10/2025

EDUCATION

Universidad Politécnica de Madrid

Master of Science, Data Science (Erasmus Semester)

Madrid, Spain

02/2026 – Present

Friedrich-Alexander-Universität Erlangen-Nürnberg

Master of Science, Data Science (Grade: 2.0)

Erlangen, Germany

10/2024 – Present

Amity University

Bachelor of Science, Mathematics (Grade: 1.8)

Noida, India

07/2020 – 06/2023

ACHIEVEMENTS

- Presented Research Paper : Delivered a presentation on "*Modelling Intuitionistic Fuzzy Sets in Real Life Using a Multi-Criterion Decision-Making Approach*" at a National Conference organized by the Department of Mathematics, LNCT Group of Colleges, Bhopal, India
- Academic Excellence : Awarded a Bronze Medal for outstanding academic performance during Bachelor of Science (Mathematics) at Amity University, Noida, India